

For the differential equation

$$x'' + bx' + 7x = 0,$$

find the values of b that make the general solution overdamped, underdamped, or critically damped.
 (For each, give an interval or intervals for b for which the equation is as indicated. Thus if the equation is overdamped for all b in the range $-1 < b \leq 1$ and $3 \leq b < \infty$, enter $(-1,1] \cup [3,\infty)$. If it is overdamped only for $b = 3$, enter $\{3\}$.)

If the equation is overdamped, $b \in$ $\mathbb{R}$.

If the equation is underdamped, $b \in$ $\mathbb{R}$.

If the equation is critically damped, $b \in$ $\mathbb{R}$.

If the differential equation

$$m \frac{d^2 x}{dt^2} + 2 \frac{dx}{dt} + 6x = 0$$

is overdamped, the range of values for m is? $\mathbb{R}$.

Your answer will be an interval of numbers given in the form $(1,2)$, $[1,2)$, $(-\infty,6]$, etc.

$$(uv)' = u'v + uv'$$

$$\frac{d}{dx}(5x^2 y) = 10xy + 5x^2 \frac{dy}{dx}$$

$$\frac{d}{dx}(u(x)y) = u'(x)y + u(x)\frac{dy}{dx}$$

1st order Linear ODE

2. To solve the first-order differential equation $t^2 y' + 2ty = \sin(t)$ using the integrating factor method, what is the required first step?

- Compute the integrating factor $\mu(t) = e^{\int 2tdt}$.
- Integrate both sides with respect to t .
- Divide the entire equation by t^2 to write it in standard form.
- Multiply the entire equation by e^{2t} .

$$t^2 y' + 2ty = \sin t$$

$$\Rightarrow y' + \frac{2}{t}y = \frac{\sin t}{t^2}$$

Integrating factor standard form

$$M[y' + p(x)y = q(x)]$$

$$\rightarrow My' + \underbrace{Mp(x)y} = Mq(x)$$

$$\rightarrow M \frac{dy}{dx} + \underbrace{M'(x)y}$$

$$\int (My)' dx = \int Mq(x) dx$$

$$My = \int Mq(x) dx$$

$$y = \frac{1}{M} \left[\int Mq(x) dx + C \right]$$

$$\frac{dM}{dx} = M' = Mp(x)$$

$$\Leftrightarrow \int \frac{1}{M} dM = \int p(x) dx$$

$$\ln|M| = \int p(x) dx$$

$$M = e^{\int p(x) dx}$$

$$(x+y)^2 = (x+y)(x+y) = x^2 + xy + yx + y^2 = x^2 + 2xy + y^2$$

$$x^2 + 18x = 3$$

$$x^2 + 18x + \left(\frac{18}{2}\right)^2 = 3 + \left(\frac{18}{2}\right)^2$$

$$(x+9)^2 = 84$$

$$x+9 = \pm\sqrt{84}$$

$$x = -9 \pm \sqrt{84}$$

$$x^3 y = 2x$$

$$(3x^2 y + x^3 \frac{dy}{dx}) = 2$$

$$\frac{x^3 \frac{dy}{dx} + 3x^2 y = 2}{x^3}$$

Solve $\rightarrow \frac{dy}{dx} + \frac{3}{x}y = \frac{2}{x^3}$

$$\frac{dy}{dx} + p(x)y = q(x)$$

$$M = e^{\int p(x) dx} = e^{\int \frac{3}{x} dx} = e^{3 \ln|x|}$$

$$= e^{\ln(x^3)} = x^3$$

$$x^3 \left(\frac{dy}{dx} + \frac{3}{x}y = \frac{2}{x^3} \right)$$

$$x^3 \frac{dy}{dx} + 3x^2 y = 2$$

$$\int (x^3 y)' dx = \int 2 dx$$

$$x^3 y = 2x + C$$

$$y = \frac{1}{x^3} [2x + C]$$

$$\frac{d}{dx} x^3 = 3x^2$$

$$\frac{d}{dx} y^3 = 3y^2 \frac{dy}{dx}$$

$$\frac{2}{2x} y^3 = 0$$

Say $y = 2x^2 + \sin x$

$$\frac{d}{dx} (2x^2 + \sin x)^3 = 3(2x^2 + \sin x)^2 (4x + \cos x)$$

$$\frac{d}{dx} y^3 = 3y^2 \left(\frac{dy}{dx} \right)$$

Example 1 Find the solution to the following differential equation.

$$\frac{dv}{dt} = 9.8 - 0.196v$$

① Notice this is a 1st order linear ODE

$$\rightarrow \frac{dv}{dt} + 0.196v = 9.8$$

$$p(x) = 0.196$$

$$q(x) = 9.8$$

② Check if it's separable — if so, use sep. of var.

$$\mu = e^{\int 0.196 dx} = e^{0.196x}$$

③ If not, then write in standard form

$$\left[\frac{dy}{dx} + p(x)y = q(x) \right]$$

④ Compute $\mu = e^{\int p(x) dx}$

⑤ The solution is

$$y = \frac{1}{\mu} \left[\int \mu \cdot q(x) dx + C \right]$$

$$v = \frac{1}{e^{0.196t}} \left[\int \underset{\substack{\uparrow \\ \mu}}{e^{0.196t}} \underset{\substack{\uparrow \\ q}}{(9.8) dt} + C \right]$$

$$\therefore v = \frac{1}{e^{0.196t}} \left[\frac{9.8}{0.196} e^{0.196t} + C \right]$$

$$\int e^{at} dt = \frac{1}{a} e^{at} + C$$

seperable
 $f(y) dy = g(x) dx$

$dy - 4y dx = e^{5x} dx$
 $\frac{dy}{dx} = 4y + e^{5x}$

1. $\frac{dy}{dx} - 4y = e^{5x}$
 $y = e^{5x} + Ce^{4x}$

2. $\frac{dy}{dx} + 3x^2 y = x^2$
 $\frac{1}{3} + Ce^{-x^3}$

3. $y' = x - 2y$
 $Ce^{-2x} + \frac{1}{2}x - \frac{1}{4}$

1st order
 Linear ✓
 Not sep. ✓
 standard form
 $y' + py = q$

$\frac{dy}{dx} + 2y = x$

$q = x$
 $p = 2$

$\mu = e^{\int p dx} = e^{\int 2 dx}$
 $\mu = e^{2x}$

$y = \frac{1}{\mu} \left[\int \mu q dx + c \right]$

$y = \frac{1}{e^{2x}} \left[\int e^{2x} x dx + c \right]$

$u = \text{Liave}$
 $\int u dv = uv - \int v du$
 $u = x \quad dv = e^{2x}$
 $du = dx \quad v = \frac{1}{2} e^{2x}$

$y = \frac{1}{e^{2x}} \left[\frac{1}{2} x e^{2x} - \int \frac{1}{2} e^{2x} dx + c \right]$

$y = \frac{1}{2} x - \frac{1}{4} + \frac{c}{e^{2x}}$

1st order Linear ODE ✓
 Not Seperable
 Standard form ✓

$p(x) = -4$
 $q(x) = e^{5x}$

$\mu = e^{\int p(x) dx}$
 $\mu = e^{-4x}$

$y = \frac{1}{\mu} \left[\int \mu q dx + c \right]$
 $= \frac{1}{e^{-4x}} \left[\int e^{-4x} e^{5x} dx + c \right]$
 $= e^{4x} \left[\int e^x dx + c \right]$
 $= e^{4x} [e^x + c]$

$y = e^{5x} + ce^{4x}$